

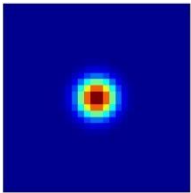
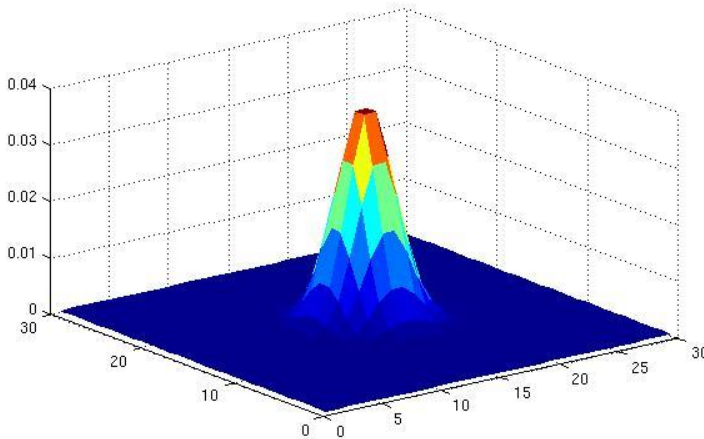


L2 Review Questions

Q1: Gaussian support

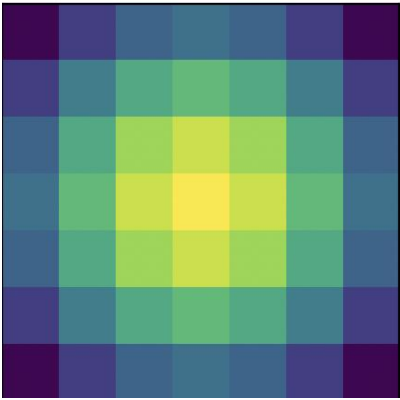
A Gaussian filter has $\sigma = 3$ and a 7×7 kernel. The kernel is changed to 15×15 while σ remains 3, and both kernels are normalized to sum to one. Which statement is most accurate?

- A. The two filters must produce identical outputs because σ is unchanged.
- B. The larger kernel better captures the Gaussian tails; once the support is sufficiently large, further increases have little effect.
- C. The larger kernel has a larger effective σ because kernel size determines the Gaussian variance.
- D. The larger kernel produces less smoothing because its weights are distributed over more pixels.

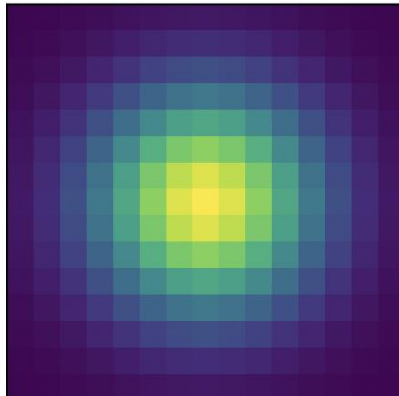


A1: Gaussian support (B)

7×7 heatmap



15×15 heatmap



- A. Same σ means the underlying Gaussian shape is the same, but a 7×7 kernel truncates more of the tails than a 15×15 kernel.
- B. larger kernel includes more of the Gaussian's tails. For $\sigma=3$, a 15×15 kernel gives substantially wider support than 7×7 .**
- C. σ determines the Gaussian variance. Kernel size determines only how much of that Gaussian is sampled/truncated.
- D. A larger kernel does not inherently mean less smoothing. Since σ stays at 3, the smoothing scale is essentially the same.

Q2: Stretching versus equalization

Images A and B have the same minimum and maximum intensity values but different intensity distributions. Histogram stretching and histogram equalization are applied independently to each image. Which statement is correct?

- A. Both operations apply the same mapping to A and B.
- B. Both operations generally apply different mappings to A and B.
- C. Equalization applies the same mapping to A and B, but stretching generally applies different mappings.
- D. Stretching applies the same mapping to A and B, but equalization generally applies different mappings.

A2: Stretching versus equalization (D)

- A. Histogram stretching would use the same mapping because both images have the same minimum and maximum intensities. But histogram equalization depends on the full histogram.
- B. Equalization generally does differ, but stretching does not in this case because the endpoints are the same.
- C. This is essentially reversed. Equalization depends on how frequently each intensity occurs, so different histograms usually mean different mappings.
- D. Same min/max $\rightarrow$ same stretching transformation. Different histograms $\rightarrow$ generally different equalization transformations.**

Q3: Normalized cross-correlation

Let T be a non-constant template, and suppose an image patch is given by:

$$P = aT + b, \quad \text{where } a > 0$$

Assuming exact arithmetic and zero-mean, unit-norm normalized cross-correlation, which statement is correct?

- A. The normalized score is 1, while the raw correlation score can change with both a and b .
- B. The normalized score is a , while the raw correlation score is unaffected by b .
- C. The normalized score is 0 because subtracting the mean removes the matching pattern.
- D. The normalized score depends only on b , while the raw correlation score depends only on a .

A3: Normalized cross-correlation (A)

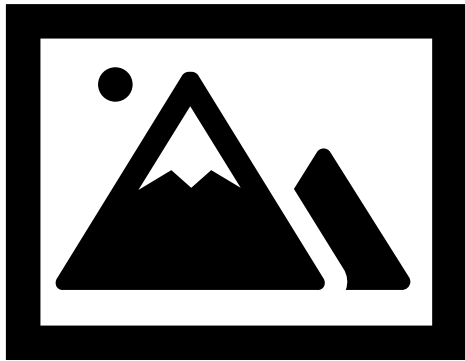
Let $\bar{T}$ and $\bar{P}$ denote the mean intensities of T and P $\bar{P} = a\bar{T} + b$.

$$P_0 = P - \bar{P} = aT + b - (a\bar{T} + b) = a(T - \bar{T}), T_0 = (T - \bar{T})$$

$$\begin{array}{l|l} \text{NCC}(T, P) = \frac{\langle T - \bar{T}, a(T - \bar{T}) \rangle}{\|T - \bar{T}\| \|a(T - \bar{T})\|} & CC(T, P) = \langle T, P \rangle = \langle T, aT + b \rangle \\ \text{NCC}(T, P) = \frac{a\|T - \bar{T}\|^2}{a\|T - \bar{T}\|^2} = 1 & CC(T, P) = a \sum_i T_i^2 + b \sum_i T_i \end{array}$$

A. The normalized score is 1, while the raw correlation score can change with both a and b .

Q4: Convolution Output Sizes



$$C = 3 \ H = 256 \ W = 256$$

An input image has shape $(H, W, C) = (256, 256, 3)$.

We apply **one 5-by-5-by-3 filter** with stride 2 and zero-padding of 2 pixels on each spatial side. What is the output shape $(H_{out}, W_{out}, C_{out})$?

- A. $(256, 256, 1)$
- B. $(128, 128, 3)$
- C. $(128, 128, 1)$
- D. $(126, 126, 1)$

A4: Convolution Output Sizes (C)

An input image has shape $(H, W, C) = (256, 256, 3)$.

$$S = 2$$

$$K = 5$$

$$P = 2$$

$$n_{out} = \left\lfloor \frac{n_{in} + 2p - k}{s} \right\rfloor + 1 \quad n_{out} = \left\lfloor \frac{256 + 2 * 2 - 5}{2} \right\rfloor + 1$$

$$n_{out} = 128$$

C. (128, 128, 1), C_out is 1 as the filter has a depth of 3 (same as the input)